Physical Cume
February 20, 2009

A metal block was heated up from 300 K to 350 K without any expansion under 1 bar pressure. The C_v for the metal piece in this temperature range is known to be 75 J/K. For this process,

(1) (20 points) Please calculate ΔU
(2) (30 points) Please calculate ΔH
(3) (40 points) Please calculate ΔS
(4) (10 points) Please calculate ΔA (Helmholtz free energy change)

For all calculations, it is necessary to show the equations and calculation procedures.